aerospace

MDPI

Review

Photometric Characterization of Space Objects: From Classical BRDF Models to Data-Driven Prediction

Liu Yang, Can Xu and Yasheng Zhang *

Graduate School, Space Engineering University, Beijing 101416, China; bananapancakeliu@outlook.com (L.Y.); 13466509187@139.com (C.X.)
* Correspondence: zhangyspublic@163.com

Abstract

The rapid proliferation of resident space objects has made space situational awareness critically dependent on accurate characterization of non-cooperative targets using photometric light curves. This review provides a comprehensive examination of data-driven approaches for space object photometric prediction, synthesizing research across optical scattering characterization, shape and attitude inversion methodologies, and intelligent analysis techniques based on machine learning and deep learning. The evolution from traditional physics-based models to contemporary data-driven paradigms is systematically analyzed, revealing fundamental trade-offs between physical interpretability, computational efficiency, and predictive accuracy. Key findings indicate that while physical bidirectional reflectance distribution function (BRDF) models provide rigorous foundations, their computational demands and prior knowledge requirements limit operational applicability; conversely, deep learning has demonstrated superior predictive accuracy in existing comparative studies, although this conclusion is qualified by the absence of standardized public benchmarks, and it also suffers from interpretability deficits and simulation-to-reality generalization gaps. Critical research gaps are identified, including the absence of public benchmark datasets, inadequate handling of temporal multi-scale phenomena, and the persistent challenge of bridging simulated and real-world observations. Future directions should pursue physics-guided machine learning frameworks that integrate domain knowledge with data-driven capabilities, develop explainable artificial intelligence techniques tailored for photometric analysis, and establish standardized evaluation protocols to advance next-generation space object characterization essential for collision avoidance and space traffic management.

Keywords: space situational awareness; photometric prediction; light curves; bidirectional reflectance distribution function; machine learning; deep learning

Academic Editor: Juan Miguel Sánchez-Lozano

Received: 19 March 2026
Revised: 26 April 2026
Accepted: 27 April 2026
Published: 29 April 2026

1. Introduction

The accelerating pace of space exploration and the burgeoning frequency of space activities have fundamentally transformed the orbital environment [1]. With thousands of active satellites, decommissioned spacecraft, and countless debris fragments populating Earth's orbits, space situational awareness (SSA) has become indispensable for safeguarding space assets and ensuring the long-term sustainability of space operations. The ability to monitor, track, and characterize resident space objects (RSOs) is no longer merely an operational convenience but a strategic necessity for collision avoidance, anomaly detection, and national security.

 https://doi.org/10.3390/aerospace13050418

Ground-based optical telescopes constitute a cornerstone of contemporary SSA infrastructure, performing critical functions ranging from maintaining and updating space object catalogs [2] to tracking anomalous satellite behaviors [3] and collecting photometric light curves [4]. Unlike radar systems that excel at detecting and tracking objects regardless of illumination conditions, optical observations provide unique insights into the physical properties of RSOs. For high-value targets in geostationary Earth orbit (GEO), where high-resolution imaging remains challenging due to vast distances, photometric data often represents the primary source of information for object characterization.

Light curves—time-series measurements of an object's apparent brightness—encode a wealth of information about the target's physical characteristics. Temporal variations in reflected sunlight are governed by a complex interplay of factors: the object's three-dimensional geometry, the reflective properties of its constituent materials, its attitude and rotational dynamics, and the evolving observational geometry relative to the Sun and the observer [3]. Consequently, the inversion and analysis of photometric data enable the inference of critical information about non-cooperative targets, including physical characteristics, identity discrimination, motion state estimation, and even operational intent assessment [3]. These insights directly support orbit prediction refinement, collision warning systems, and on-orbit servicing operations.

The optical cross-section (OCS) serves as the fundamental metric for quantifying an object's photometric signature. Defined as the effective scattering area of a target, OCS depends exclusively on surface material properties, geometric structure and dimensions, attitude, and the directions of illumination and observation, while remaining independent of detection distance and specific system parameters [5]. This invariance makes OCS an ideal quantity for establishing mappings between observable photometric signals and intrinsic target characteristics.

Despite the wealth of information contained within photometric data, significant challenges persist in extracting actionable intelligence. Under favorable conditions, a single ground-based optical telescope can observe hundreds of objects per night, generating massive volumes of photometric time series. However, current utilization of this data remains largely confined to one-dimensional temporal analysis, often neglecting the crucial relationship between photometric signatures and observational geometry. This represents a substantial untapped potential for enhancing space object characterization capabilities.

The traditional paradigm for photometric prediction follows a "model-to-data" workflow: target modeling, determination of observational geometry, and photometric calculation. While this approach has enabled estimate shape and posture estimation from observational data, it imposes stringent requirements on observational geometry and suffers from accumulated uncertainties along the modeling chain. Furthermore, as observational capabilities continue to advance, the resulting data deluge demands novel analytical approaches capable of mining high-value intelligence from massive photometric datasets. The conventional shape and posture inversion methods, heavily reliant on accurate prior models, struggle to maintain accuracy when such models are unavailable or uncertain.

This review addresses the urgent need for "data-to-data" photometric prediction methodologies that bypass extended forward modeling chains and directly learn mappings from observational data to target characteristics. We synthesize research across three interconnected domains: optical scattering characteristics modeling, shape and attitude inversion techniques, and intelligent analysis approaches leveraging machine learning and deep learning. By critically examining the evolution from physics-based models to data-driven paradigms, we identify persistent challenges and chart future research directions toward robust, interpretable, and generalizable photometric prediction capabilities.

https://doi.org/10.3390/aerospace13050418

It is worth noting that the majority of techniques reviewed in this paper—particularly those relying on detailed three-dimensional geometric models, high-fidelity BRDF parameterization, and deep learning with high signal-to-noise ratio light curves—are developed for space objects with characteristic dimensions on the order of $\geq$1 m (e.g., operational satellites). For small debris in the 10–30 cm size range, many of these methods face significant challenges due to rapid tumbling motion, simpler yet less predictable geometries, lower signal-to-noise ratios, and the lack of prior knowledge. While the underlying physical principles remain valid, the practical applicability of model-based inversion and intelligent classification techniques to such small debris requires further investigation and is beyond the primary scope of this review.

The remainder of this paper is organized as follows. Section 2 reviews the fundamental principles of optical scattering characterization, from microscopic BRDF models to integrated OCS computation methods. Section 3 examines shape and attitude inversion techniques, contrasting model-based and feature-based approaches. Section 4 surveys intelligent analysis methodologies, encompassing traditional machine learning and contemporary deep learning applications. Section 5 synthesizes key findings, identifies research gaps, and proposes future directions. Section 6 concludes the review.

2. Optical Scattering Characteristics of Space Objects

2.1. Bidirectional Reflectance Distribution Function: Theoretical Foundations

The quantitative description of surface scattering behavior originated with Nicodemus [6], who formally introduced the BRDF framework at the National Bureau of Standards in 1965. Rooted in geometric optics and radiometry, BRDF provides a mathematical function $f_r(\theta_i, \phi_i; \theta_r, \phi_r)$ that precisely defines the directional characteristics of surface reflection. It is defined as the ratio of the differential radiance dL_r reflected in the direction (θ_r, ϕ_r) to the differential irradiance dE_i incident from the direction (θ_i, ϕ_i):

$$f_r(\theta_i, \phi_i; \theta_r, \phi_r) = \frac{dL_r(\theta_i, \phi_i; \theta_r, \phi_r; E_i)}{dE_i(\theta_i, \phi_i)} \tag{1}$$

BRDF quantifies how incident radiant energy is redistributed across the hemisphere, capturing the continuum of reflection behaviors from nearly specular to ideally diffuse. The functional form depends critically on the surface's microscopic geometry, complex refractive index, and subsurface scattering characteristics, earning BRDF its designation as the "fingerprint" of material optical properties. Figure 1 illustrates the geometric relationship of the BRDF.

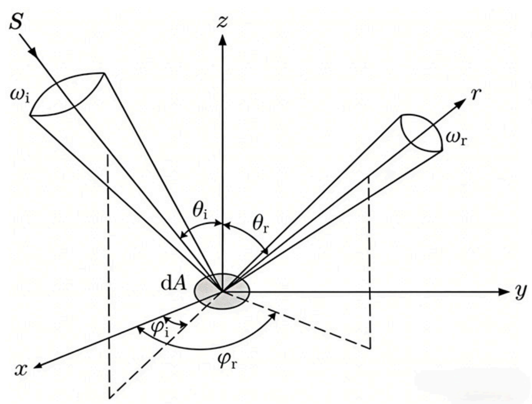

Figure 1. Geometric relationship of BRDF. The dashed lines represent the projections of the incident and reflected directions onto the surface plane (x–y plane).

2.2. Evolution of BRDF Models

BRDF model development has followed three parallel trajectories, empirical, theoretical, and experimental approaches, each offering distinct trade-offs between physical accuracy, computational efficiency, and practical applicability.

2.2.1. Empirical Models

Empirical models provide computationally efficient approximations of reflection phenomena through simplified mathematical formulations with adjustable parameters. These models prioritize speed and flexibility over strict physical fidelity, making them particularly suitable for computer graphics applications where real-time rendering is essential.

A foundational approach in empirical BRDF modeling is the decomposition of surface-reflected intensity into ambient, diffuse, and specular components, as exemplified by the Phong model [7]. Despite its widespread adoption, this model struggles to accurately capture Fresnel reflection phenomena typical of space object materials. To address this limitation, modified Phong formulations have been developed that introduce additional parameters to regulate Fresnel intensity and its angular attenuation [8]. However, these modified models still retain a relatively simplistic diffuse component and suffer from sensitivity to parameter initialization, which complicates their use in inversion tasks.

An alternative empirical strategy builds on the Torrance–Sparrow framework [9] and employs a five-parameter model optimized via genetic algorithms [10]. This approach effectively avoids local optima convergence that plagues traditional optimization methods when applied to complex nonlinear problems. Nevertheless, its reliance on empirically determined parameters—requiring extensive calibration—and the suboptimal computational efficiency of genetic algorithms, especially over broad parameter ranges, remain practical drawbacks.

2.2.2. Theoretical Models

Theoretical models derive reflection behavior from fundamental physical principles, offering superior accuracy at the cost of increased complexity and computational demands. These models are indispensable for applications requiring high-fidelity predictions.

The Torrance–Sparrow model [10] fundamentally transformed surface reflection modeling by establishing the first systematic connection between reflection and realistic surface microscopic geometry. By introducing geometric attenuation and Fresnel effects, this model achieved unprecedented realism in rendered results, particularly for grazing-angle observations. Critically, its parameters possessed physical meaning, enabling direct measurement from real materials rather than heuristic tuning. The Torrance–Sparrow model laid the foundation for modern physically based rendering, influencing subsequent developments including the renowned Cook–Torrance model [11].

Building upon this foundation, Cook and Torrance [11] at Cornell University developed their eponymous model as a comprehensive physically based BRDF framework. The Cook–Torrance model remains one of the most complete and complex BRDF formulations, applicable to diverse materials across varying roughness scales. However, despite ongoing simplifications, its computational intensity continues to limit widespread adoption in time-sensitive applications.

2.2.3. Experimental Models

Experimental models derive functional forms through analysis and fitting of measured BRDF data obtained from instrumentation. These models inherently depend on the accuracy and representativeness of experimental measurements, constraining their applicability to materials that have been empirically characterized.

Schlick's BRDF model [12] introduced a general framework that simplified computationally expensive Fresnel terms while maintaining physical plausibility. By explicitly classifying materials as single-face or double-face reflectors, the Schlick model occupied an intermediate position between purely empirical approaches and fully physical models. However, its treatment of microfacet distribution and geometric occlusion employed relatively simplified schemes compared to complete Torrance–Sparrow or Cook–Torrance formulations, resulting in some accuracy compromise.

The Lafortune model [13] adopts a fundamentally different approach, representing complex reflection phenomena by fitting basis functions to experimental data. This formulation achieves a compact representation of various reflection behaviors, including anisotropy and retro-reflection, with remarkably few parameters. While it is highly effective for rough metals and painted surfaces, its applicability diminishes for strongly diffuse materials.

Addressing the challenge of multiple scattering in complex spacecraft materials, Donnelly [14] developed a macroscopic roughness model specifically for satellite multi-layer insulation (MLI) reflectance. By incorporating surface roughness effects into the BRDF framework, this work provides a more accurate representation of MLI's reflective behavior, which is crucial for predicting the photometric signatures of spacecraft with extensive thermal blanket coverage. This approach highlights the importance of material-specific modeling in reducing simulation-to-reality discrepancies.

Table 1 presents a comparison of BRDF model types, characteristics, and representative models.

Table 1. Comparison of BRDF model types, characteristics, and applications.

Type	Characteristics and Applications	Representative Models
Empirical	Provide computationally simple formulations for specific reflection phenomena. Rapid calculation with adjustable parameters. Widely used in computer graphics simulation.	Phong model [7]; Five-parameter model [9]
Theoretical	Derived from rigorous physical laws. High accuracy with complex expressions and computational demands. Employed in precision-critical research.	Torrance–Sparrow [10]; Cook–Torrance [11]
Experimental	Derived through analysis and fitting of measured BRDF data. Strong dependence on experimental accuracy. Applicability limited to characterized materials.	Schlick model [12]; Lafortune model [13]

2.2.4. Emerging Data-Driven Approaches

Recent years have seen the emergence of neural network-based methods for material appearance modeling, offering novel possibilities for space object characterization. One such technique, neural multi-resolution material representation (NeuMIP), employs a compact multilayer perceptron (MLP) network as a universal material representation [15]. By taking texture coordinates and illumination-viewing directions as input, the network generates comprehensive material appearance spanning macroscopic color to microscopic detail, achieving hundredfold data compression while maintaining seamless multi-resolution rendering.

Extrapolating this paradigm to space object modeling holds promise for dramatically enhancing the efficiency of large-scale photometric simulations while providing high-fidelity solutions inaccessible to traditional models. However, this technology remains nascent for aerospace applications demanding exceptional reliability, facing challenges including physical consistency guarantees, scarcity of on-orbit material training data, and extrapolation reliability concerns.

A novel integration of physical BRDF models with neural representation learning has been proposed by Zhang and Rupnik [16], who developed BRDF-NeRF—a neural radiance field framework that incorporates semi-empirical RPV (Rahman–Pinty–Verstraete) BRDF modeling for optical satellite imagery. This approach enables surface reconstruction from sparse multi-angle observations while explicitly modeling anisotropic reflectance behavior, effectively combining the physical interpretability of parametric BRDF models with the representational power of neural networks. Although primarily developed for Earth observation applications, the BRDF-NeRF paradigm offers promising implications for space object characterization, suggesting a pathway toward data-driven photometric models that retain physical consistency while learning complex reflectance patterns directly from observational data.

2.3. Integrated Optical Cross-Section Computation

While BRDF models characterize material-level scattering behavior, operational SSA requires predicting the integrated photometric signature of complete space objects with complex three-dimensional configurations. This transition from local material properties to global object characteristics necessitates computation of the optical cross-section, which relates to BRDF through

$$\sigma_{\mathrm{OCS}} = \sum_i f_r(\theta_i, \phi_i; \theta_r, \phi_r) \cdot \Delta A_i \cdot \cos\theta_i \cdot \cos\theta_r \cdot V_i \tag{2}$$

where ΔA_i represents the area of surface facet i, θ_i and θ_r denote incidence and reflection angles relative to the facet normal, and Vi indicates facet visibility to both illumination source and detector [17]. A schematic diagram of the OCS variables is shown in Figure 2. As expressed, OCS fundamentally represents the integrated BRDF response of all visible surface elements weighted by their projected areas and geometric relationships.

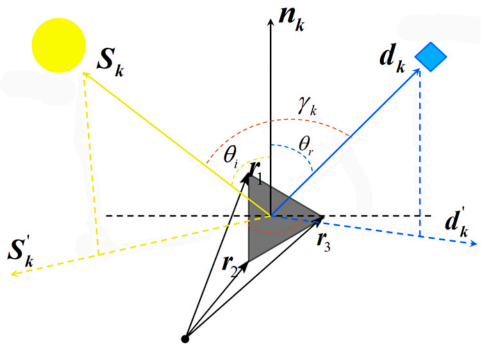

Figure 2. Schematic diagram of OCS variables. The gray shaded facet represents a surface element ΔA_i. The solid arrows indicate the incident vector S_k (yellow, from the facet to the light source) and the reflection vector d_k (blue, from the facet to the detector). The dashed arrows S_k' and d_k' denote the projections of S_k and d_k onto the local tangent plane of the facet. The vectors r_1, r_2, r_3 are the components of the position vector from the center of mass (black dot) to the facet. The angles θ_i (incidence), θ_r (reflection), and γ_k (phase angle) are defined with respect to the facet normal n_k.

Computational approaches for OCS determination have evolved along four principal trajectories: analytical and numerical methods based on physical equation solving; graphics rendering techniques leveraging computer graphics hardware acceleration; experimental validation through empirical measurement; and integrated multi-physics simulation frameworks.

2.3.1. Analytical and Numerical Methods

Traditional analytical approaches derived OCS formulas for canonical geometries through mathematical derivation from radiometric principles. Sandia National Laboratories pioneered investigations of low Earth orbit microsatellite optical scattering characteristics, deriving OCS expressions for spheres, cubes, and prisms with varying surface materials, enabling simulation of optical signatures of small satellites under different attitudes.

Surface discretization numerical integration extended analytical capabilities to complex geometries. Bao et al. [17] developed a facet-based meshing method for computing optical characteristics of complex space objects. By decomposing satellites into combinations of standard geometric primitives and discretizing continuous surfaces into finite facets, Figure 3 presents a schematic diagram of facet discretization for a space object. This approach achieved superior accuracy compared to purely analytical methods while handling complex occlusion relationships. However, neglecting multiple inter-facet reflections introduced potential errors in high-reflectance scenarios.

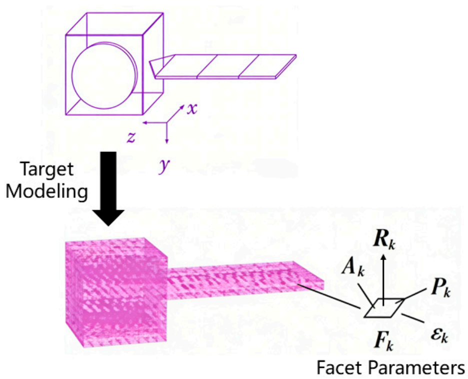

Figure 3. Schematic diagram of facet discretization for a space object. The axes x, y, z define the global coordinate system. Each facet F_k is represented by its centroid P_k, normal vector R_k, area A_k, and diffuse reflection coefficient ε_k.

2.3.2. Graphics Rendering Methods

Graphics rendering methods leverage computer graphics application programming interfaces and hardware acceleration to transform OCS computation into pixel luminance integration problems within real-time three-dimensional rendering pipelines.

A fast visualization computation framework based on Open Graphics Library and three-dimensional models has been developed for efficient OCS computation [18]. By leveraging OpenGL's capabilities in geometric transformation, occlusion determination, and multi-angle illumination simulation, combined with Phong model-based surface element optical contribution calculation, this approach achieves efficient OCS computation. Despite limitations in BRDF modeling fidelity and surface discretization treatment, its efficiency and practicality have established valuable tools for space object optical analysis.

To overcome the real-time limitations of facet-based methods and the limited BRDF description capability of traditional graphics approaches, an efficient computation method combining OpenGL picking technology with improved Z-buffer techniques was developed [19]. The core innovation uses OpenGL picking for primary occlusion determination followed by refined Z-buffer analysis for precise secondary occlusion assessment, enabling rapid extraction of visible facets. Supporting high-precision BRDF models including the five-parameter Phong formulation, this method achieves enhanced accuracy, although sensitivity to model construction and measurement positioning errors persists under extreme observational geometries.

Beyond methodological advancements in accuracy, the practical applicability of graphics rendering methods hinges critically on computational efficiency. Maxwell [20] addressed the computational bottleneck of high-fidelity photometric modeling by developing efficient algorithms for complex spacecraft geometries. Their approach leverages graphics rendering techniques to accelerate optical cross-section calculations while maintaining physical accuracy in BRDF representation. By demonstrating that careful algorithmic optimization can reduce computation time by orders of magnitude without sacrificing fidelity, this work provides a practical pathway toward integrating physically accurate photometric models into large-scale simulation frameworks and iterative inversion schemes.

2.3.3. Experimental Validation and Empirical Analysis

Experimental approaches ground computational models in physical reality through ground-based simulation experiments or astronomical observations, enabling model validation and empirical relationship development.

A critical challenge in space object optical characterization is the validation of satellite scattering simulations against real measurements. One solution to this problem is the use of ground-based experimental platforms. For instance, a high-automation platform with multi-degree-of-freedom attitude adjustment has been developed to enable precise parallel simulation of on-orbit satellite motion. By incorporating modified Phong models with surface wrinkle treatment, such platforms improve the characterization of non-Lambertian materials. This line of work has established a reliable paradigm that tightly integrates experimental measurement with simulation-based analysis [21].

Complementing these ground-based experimental efforts, empirical validation using actual telescopic observations has also been pursued. In one study, measured data from 29 Starlink satellites observed with RGB-filtered telescopes were analyzed. Precise extraction of apparent magnitudes, intensity curves, and celestial coordinates allowed for the correlation of satellite brightness with observational geometry across multiple color bands. The findings revealed wavelength-dependent reflection patterns: the reflection gain curves in the blue band exhibited a distinct convexity compared to other bands [22]. The empirical brightness-geometry relationships derived from such work provide a foundation for future space object identification and brightness prediction.

Beyond dedicated experimental setups and targeted observations, existing scientific missions can also serve as valuable sources of empirical data for space object characterization. Billot et al. [23] demonstrated that scientific missions like CHEOPS (CHaracterising ExOPlanets Satellite) can serendipitously contribute to space situational awareness. By applying Hough transform algorithms to the mission's science images, they detected thousands of RSO trails, revealing increased occurrence rates over time and the distinctive signature of the Starlink constellation. This highlights the potential of leveraging existing astronomical infrastructure for supplementary RSO characterization.

Beyond ground-based facilities and repurposed scientific missions, stratospheric balloon platforms offer a cost-effective intermediate tier for validation under near-space conditions. Suthakar et al. [24] presented RSONAR (Resident Space Object Network for Assessment and Recognition), a data-driven evaluation framework using a dual-use star tracker mounted on a stratospheric balloon platform for space situational awareness validation. Their high-altitude experiments demonstrating the feasibility of low-cost platforms for RSO characterization. By developing algorithms to extract light curves and classify objects from star tracker imagery, this work provides a real-world testbed for validating simulation-based models and bridges the gap between laboratory simulations and operational observations. Such experimental frameworks complement ground-based and space-based validation efforts, collectively enabling multi-tiered empirical assessment of optical characterization methodologies.

2.3.4. Integrated Simulation Frameworks

Comprehensive simulation frameworks integrate orbital mechanics, attitude dynamics, material properties, and environmental factors into end-to-end models enabling high-fidelity optical predictions.

Du [25] systematically constructed a complete computational framework encompassing radiometric theory, satellite orbit and attitude determination, and surface material optical characterization for on-orbit satellite visible light scattering. The component-based modeling approach captured complex satellite structures, while integration with Satellite Tool Kit [26] enabled realistic simulation of actual orbital and attitude dynamics. This work quantitatively revealed how orbital configuration, flight attitude, and component materials influence scattering characteristics, though geometric model simplification presented optimization opportunities.

Efforts to achieve high-fidelity end-to-end simulation have also led to the development of autonomous imaging simulation frameworks. One such framework integrates detailed three-dimensional modeling, precise orbit and attitude determination, and OpenGL-based geometric projection with complete imaging chain modeling. By comprehensively accounting for target surface BRDF scattering, atmospheric turbulence, optical system modulation transfer function, and detector noise, this platform enables dynamic simulation of on-orbit target optical image sequences. It has been shown to offer advantages in modeling fidelity for certain spacecraft, thereby supporting on-orbit status monitoring and non-cooperative target attitude recognition [27].

Another line of work focuses on predicting ground-observable flash sequences generated by specular panel reflection. A light curve model specifically designed for this purpose incorporates mirror curvature effects through discretized normal vectors, which refines the characterization of reflection geometry [28]. This approach addresses the flash gaps inherent in planar mirror models and explains transitions between mirror groups. It has been successfully applied to precise satellite rotation parameter determination, though validation has primarily relied on data from Japan's Ajisai geodesy satellite, leaving broader applicability to be confirmed.

Beyond analytic BRDF formulations, emerging data-driven approaches offer new possibilities for material modeling in space object simulations. Li et al. [29] proposed PureSample, a novel neural BRDF representation that learns material behavior purely by sampling forward random walks on explicit microgeometry—eliminating the need for analytic BRDF derivation. By employing a flow matching network for importance sampling and a lightweight neural network for view-dependent albedo, this framework enables efficient BRDF evaluation and sampling for complex materials including multi-layered and multiple-scattering microstructures. While developed for computer graphics, this approach offers promising implications for space object material modeling, suggesting a pathway toward data-driven BRDF representations that learn directly from physical scattering simulations rather than relying on simplified analytic approximations. The integration of such neural representations into future simulation frameworks could significantly enhance their fidelity for materials where analytic models fall short.

2.4. Summary and Research Gaps

Despite significant advances in BRDF modeling and OCS computation, persistent challenges impede progress toward SSA requirements demanding high precision, real-time responsiveness, and operational robustness.

Accuracy-Efficiency Trade-off: High-fidelity physical models such as Cook–Torrance achieve superior accuracy but incur computational costs prohibitive for integration with time-sensitive orbital and attitude dynamics simulations. Conversely, efficient empirical or experimental models like the five-parameter Phong formulation enable rapid computation but exhibit limited extrapolation capability for novel materials or extreme observational geometries. Current research lacks systematic guidance for model selection optimized to specific application requirements, balancing accuracy and efficiency trade-offs.

Complex Target and Environment Modeling: Multiple scattering between facets can contribute significantly under specific observational geometries, yet prevailing facet-based or graphics methods typically consider only single scattering or employ severely simplified multiple scattering approximations, introducing non-negligible errors. Furthermore, existing models universally assume constant surface BRDF, neglecting space environmental degradation effects including atomic oxygen erosion, ultraviolet radiation damage, and micrometeoroid impacts. This time-varying behavior compromises long-term photometric prediction accuracy, yet current models lack effective integration of these dynamic processes.

Prior Information Dependency: Many OCS computation methods require precise prior three-dimensional geometric models, rendering them problematic for non-cooperative targets whose detailed structures remain unknown. This dependency fundamentally limits applicability in realistic operational scenarios where targets may be encountered without prior knowledge.

3. Shape and Attitude Inversion from Photometric Data

3.1. Model-Based Inversion Approaches

Model-based inversion techniques constitute a core methodology in contemporary SSA, inferring target characteristics by matching theoretical predictions from established models against observational data. These approaches establish physical models encompassing target geometry, surface optics, and attitude dynamics, then optimize model parameters to minimize discrepancies between simulated and measured light curves [30]. The fundamental paradigm is inherently "model-driven," with reliability critically dependent on the fidelity with which prior models approximate real target properties.

The typical workflow proceeds through sequential stages [31,32]:

- Geometric Modeling: Construction of a three-dimensional geometric model of the target based on prior knowledge.
- Optical Parameter Assignment: Assignment of preliminary optical parameters (e.g., BRDF) to the constituent components of the model.
- Attitude Dynamics Modeling: Development of attitude evolution models that incorporate orbital mechanics and attitude dynamics.
- Photometric Simulation: Simulation of theoretical light curves under specified observational geometries using ray-tracing or graphics rendering techniques.
- Iterative Parameter Optimization: Optimization of attitude parameters, angular velocities, and surface properties through iterative algorithms [31,32] to minimize the discrepancy between the simulated and measured light curves.

This physically grounded approach offers interpretable inversion results and has become indispensable for decoding photometric signals and enabling non-cooperative target characterization. Research evolution reveals a clear trajectory: from simple state estimation under strong assumptions, through refined physical modeling and parameter inversion, to systematic methodology evaluation and validation, culminating in common platform development supporting future research.

Early investigations approached shape and attitude inversion as state estimation problems requiring dynamical priors. Shan et al. [33] proposed joint attitude and angular velocity estimation from time-series photometric observations. By comprehensively modeling Sun–Earth–target geometric relationships, target shape, and Phong model-based surface reflection characteristics affecting observed magnitudes, they constructed complete photometric observation models. The geometric relationship of space-based observation is depicted in Figure 4. Integration with modified Rodrigues parameter-described attitude and angular velocity kinematics enabled unscented Kalman filter [31] processing for online joint estimation of unknown initial attitude and angular velocity, with adaptive tracking capability for slow maneuvers. While such methods excel at time-series data processing and continuous state estimation, they require reasonable initial values and depend heavily on dynamical model accuracy.

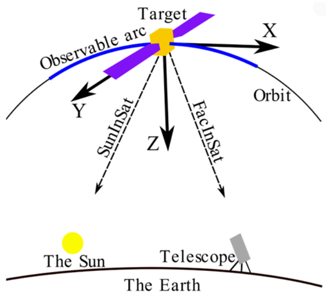

Figure 4. Geometric relationship of space-based observation. Gray indicates the orbit, and blue indicates the observable arc.

Addressing the critical gap between high-fidelity inversion and real-time operational requirements, Renis [34] developed a framework that leverages light curve inversion to enhance kinematic state estimation of uncooperative targets. By extracting motion priors from photometric data and integrating them into an unscented Kalman filter architecture, this approach achieves improved tracking performance without sacrificing computational efficiency. Validation using high-fidelity VESPA payload data demonstrates that incorporating inversion-derived information significantly reduces state estimation uncertainty compared to filter-only methods. This work demonstrating how physically meaningful information extracted from light curves can augment conventional tracking algorithms in operational settings.

Recognizing that dynamical models often remain unknown or inaccurate for non-cooperative targets, subsequent research shifted focus directly to the physics of light scattering, treating inversion as optimization problems. By constructing parameterized photometric models and adjusting parameters (attitude, surface optical properties) to optimize the matching between simulation and measurement, these approaches reduced the requirements for dynamical information and improved application flexibility.

Addressing precessing target motion state inversion challenges, Tian et al. [35] developed OCS timing analysis methods employing modified Phong models with surface wrinkle treatment for improved OCS accuracy. Satellite tool kit (STK) simulation generated OCS sequences under varying precession angles, with cross-residual analysis revealing periodic characteristics. Systematic identification of precession angle effects on optical scattering periodicity demonstrated that precession angles below 30° enable effective rotation rate inversion, while exceeding this threshold eliminates periodicity—providing theoretical foundations for identifying uncontrolled satellite states. However, conclusions derived from specific satellite model simulations warrant broader validation in real complex space environments.

Recognizing limitations of traditional two-surface models [30] in capturing the inherent specular reflection (satellite glints), Zhang et al. [36] developed multi-component models introducing specular reflection elements as third components alongside diffuse bodies. Complete parameter inversion methodologies included determining diffuse reflectance through photometric curve fitting, resolving specular component spatial orientation using reflection principles and satellite motion geometry via Euler angle methods [36], and computing effective areas through glint peak magnitude comparison—successfully reproducing glint phenomena observed in measured data.

Similar objectives have been pursued using specular geometry optimization, where adjusting mirror longitude configurations significantly enhances model fit [28]. High-frequency, high-temporal-resolution photometric data—such as those acquired with France's Grasse Observatory M60 telescope—enable direct inversion of satellite spin axis direction, rotation period, and initial rotation angle, achieving high-precision rotation state determination. Nevertheless, the accuracy of such inversions remains partially constrained by limited prior knowledge of the satellite surface's specular geometry distribution.

A fundamental challenge of light curve inversion is its ill-posed nature. To address this, a physically grounded forward model based on the Cook–Torrance BRDF formulation has been developed, which leverages multi-colour observations to simultaneously constrain attitude and material parameter estimation [37]. By exploiting Johnson–Cousins multi-colour photometry, this approach couples wavelength-invariant geometric parameters with wavelength-dependent reflectance properties, thereby narrowing the admissible solution space compared to traditional panchromatic inversions. This work demonstrates how multi-spectral information can regularize the inversion process and mitigate the risk of non-unique or erroneous parameter estimates. The integration of multi-colour constraints thus offers a promising pathway for enhancing the robustness of model-based inversion methods, particularly for non-cooperative targets where prior information is limited.

Different target classes with distinctive observational characteristics require specialized inversion treatments. For attitude monitoring of three-axis stabilised satellites—where targets maintain nominally fixed orientations and traditional inversion methods often struggle due to limited signal modulation—a dedicated high-performance light curve inversion method has been developed [38]. By incorporating robust optimization techniques, this method extracts subtle attitude variations from photometric data with improved convergence properties and reduced sensitivity to observational noise. Such advances enhance the operational applicability of model-based inversion for active satellite monitoring.

As physical model diversity increased, systematic evaluation frameworks emerged to assess comparative performance across inversion accuracy, computational efficiency, and convergence characteristics, providing scientific foundations for model selection. Clark et al. [32] systematically compared two BRDF models—standard facet-based and anisotropic Phong formulations—under simulated conditions using particle swarm optimizers for parameter estimation. Despite Phong's superior physical fidelity, the facet-based model demonstrated excellent comprehensive performance due to fewer parameters and simpler response surfaces, achieving higher pose estimation accuracy, faster computation, and improved convergence. This work provides methodological foundations and model selection guidance for low-cost sensor-based RSO characterization within widely distributed low Earth orbit (LEO) constellations.

All of the above inversion methods share a common bottleneck: insufficient high-fidelity real measurement data for development and validation. Consequently, cutting-edge research focuses on digital twins or high-fidelity simulation platforms that create highly realistic virtual environments simulating the complete chains from targets, orbits, attitudes, illumination, to sensor noise—providing reliable training data and performance benchmarks for inversion methodologies.

Robinson and Frueh [39] recently proposed a comprehensive digital twin model for optical ground station image acquisition in SSA applications, enabling high-precision CCD/CMOS sensor image simulation. Integrating multi-level space object brightness models, precise Gaia DR3 star catalog-based stellar signals, and multiple background noise sources including airglow, moonlight scattering, and light pollution with sensor noise, validation against real observations from Purdue University Optical Ground Station and Switzerland's ZimMAIN telescope demonstrated highly realistic image generation when system parameters are well-characterized. This digital twin supports telescope system design, observation mission planning, and machine learning training data generation.

3.2. Feature-Based Inversion Approaches

In contrast to model-based methods, feature-based inversion approaches follow "data-driven" trajectories, extracting characteristics directly from photometric curves for state parameter estimation without relying on precise prior three-dimensional models.

Light curve inversion for estimating celestial body shape and attitude has evolved over more than a century, initially focused on asteroids and natural satellites. Such studies typically invoked reasonable simplifying assumptions: stable single-axis rotation and approximately smooth diffuse surfaces [40]. However, transitioning to near-Earth artificial satellites fundamentally challenges these assumptions. Unlike natural bodies, satellites exhibit qualitatively different characteristics: surfaces composed of diverse materials with distinct reflection properties, angular geometries rather than streamlined forms, and complex mixed specular-diffuse reflection behaviors. Attitude motions grow correspondingly complex; even initially spin-stabilized satellites experience rotation rate decay due to eddy currents, while solar radiation pressure and shadow effects ultimately induce unstable attitude evolution [41]. Consequently, space object light curve inversion complexity dramatically exceeds that of natural bodies.

Nevertheless, feature-based inversion offers decisive advantages: even without prior target knowledge, direct photometric curve analysis enables platform type identification, angular velocity estimation, and spin period determination [30]. This rapid characterization capability supports efficient target classification and anomaly detection while providing valuable initial parameters for more complex model-driven inversion.

Feature-based inversion follows three principal technical trajectories:

- Period and rate inversion based on traditional signal processing;
- Attitude trajectory inversion combining physical models with intelligent optimization;
- Attitude pattern recognition through machine learning feature extraction.

Traditional signal processing approaches treat light curves as one-dimensional time series, applying classical techniques for periodic feature extraction and simple parameter inversion. Tian et al. [42] systematically applied spectral analysis, autocorrelation, and cross-residual methods to experimental photometric data for unstable target rotation rate inversion, demonstrating that cross-residual analysis most effectively suppresses multi-surface flasher interference, achieving the highest rotation rate accuracy. This provides reliable technical pathways for rotation parameter extraction in space object status monitoring. However, experimental simplifications considering only single-axis rotation failed to simulate complex multi-axis tumbling motions.

Physical model-based approaches with intelligent optimization transcend single-parameter extraction to recover complete attitude evolution histories. The core "simulation-matching" paradigm establishes physical models incorporating attitude dynamics and light scattering, then adjusts attitude parameters through intelligent optimization to match simulated and measured curves optimally. Burton et al. [40] innovatively proposed two-stage particle swarm optimization: first-stage non-social particle swarm optimization (PSO) independently searches possible attitudes at each observation epoch, identifying multiple local optima; second-stage full-state PSO combines non-social PSO (nsPSO) outputs with analytical torque-free attitude motion models including short-axis mode and long-axis mode [43] for rapid attitude history propagation and complete curve matching. By incorporating analytical attitude expressions, this approach avoids numerical integration computational overhead, proving particularly suitable for fast-tumbling object attitude estimation. However, accuracy remains critically dependent on physical model accuracy; performance degrades when prior information is uncertain or unavailable.

Machine learning-based pattern recognition represents the most thoroughly "data-driven" trajectory, attempting to bypass precise physical modeling by learning direct mappings from photometric curve morphological features to target attitude patterns. Isoletta et al. [44] proposed attitude classification methods requiring no prior target shape, orbit, or surface reflection information. Lomb–Scargle periodogram [45] analysis extracts spectral features, classifying attitudes through peak frequencies and periodic characteristics, while phase dispersion minimization [46] tests candidate rotation periods for final attitude type determination. Despite sensitivity to high spin rates, target shape, and observation duration, this work fundamentally shifts feature-based inversion from "physical model-driven" to "data feature-driven" paradigms, with revolutionary advantages in prior information independence.

Extending this paradigm shift, Bencivenga and Isoletta [47] further refined their approach with comprehensive validation on publicly available datasets. Their framework applies Lomb–Scargle periodogram analysis to extract spectral features and phase dispersion minimization for candidate period testing, successfully discriminating between different attitude regimes—including stable rotation, precession, and tumbling—using only photometric measurements. This work consolidates the transition to data feature-driven paradigms and demonstrates the practical viability of prior-free attitude characterization for rapid space object assessment.

Building on the exploration of inversion-based methods for space object characterization, recent studies have further probed both the potential and the limitations of recovering physical attributes from photometric data. Bhowmick and Kumaran [48] used deep neural networks to investigate how much shape information is embedded in light curves. Their results show that shape complexity parameters can be recovered with errors below 15%, but retrieval accuracy degrades for highly complex shapes ($C > 0.3$), quantitatively revealing the fundamental limits of feature-based inversion approaches.

A critical challenge in supervised learning for attitude estimation is the limited availability of labeled data. To overcome this, unsupervised metric learning has been explored. One such method learns an embedding space where light curves from the same attitude state are pulled together while those from different states are pushed apart—without requiring ground-truth attitude labels [49]. This enables attitude state discrimination and change detection purely from observational data. The work demonstrates the potential of self-supervised representation learning for extracting physically meaningful information from unlabeled photometric measurements, complementing the supervised paradigms explored above.

Beyond algorithm development, the adequacy of single-source photometric data for shape and spin recovery has also been evaluated. Using multi-epoch observations from the Transiting Exoplanet Survey Satellite (TESS) and light curve inversion with Lommel–Seeliger ellipsoids, a systematic study of 44 main belt asteroids showed that while TESS-only data performs well for determining rotation periods, it struggles with detailed shape and spin-axis orientation recovery—achieving only moderate agreement with the Database of Asteroid Models from Inversion Techniques (DAMIT) [50]. This finding carries an important lesson for artificial space object characterization: single-source photometric data may suffice for bulk parameter estimation such as rotation rate, but accurate shape reconstruction likely requires multi-view or multi-epoch observations covering diverse viewing geometries.

The rotational motion of defunct space objects can also be characterized through feature-based analysis of light curves. For example, a parametric study correlating simulated predictions with photometric observations from the Falcon Telescope Network investigated the rotational motion of defunct rocket bodies [51]. The results revealed that the length-to-diameter ratio significantly influences nutation and precession behavior, manifesting as distinct oscillation patterns in light curves.

Building on this understanding of the information that can be extracted, recent work has focused on developing architectures capable of full three-dimensional reconstruction. Tang et al. [52] developed a deep learning framework that directly maps light curves to three-dimensional shape representations using convolutional neural network (CNN)–transformer architectures. Their approach achieves millisecond-level inversion speed and accurately predicts concave regions with an intersection over union (IoU) of 0.89. IoU is a common metric in computer vision that measures the overlap between predicted and ground truth regions, defined as the area of their intersection divided by the area of their union; a value of 1 indicates perfect agreement. This result demonstrates that neural networks can overcome the computational bottlenecks of traditional optimization methods while preserving shape reconstruction fidelity.

Beyond advancing reconstruction capabilities, the ultimate value of feature-based inversion lies in its integration into operational space situational awareness systems. Tsaprailis et al. [53] developed methods for simultaneous spin status determination and initial orbit determination from single optical tracks. By combining feature extraction from photometric data with orbital propagation constraints, their approach enables rapid characterization of newly detected objects without requiring multiple observation passes. This work represents a significant step toward embedding data-driven methods directly into operational SSA workflows, addressing the transition from research concepts to practical deployment.

3.3. Summary and Remaining Challenges

Examining existing research trajectories reveals both technical progress and fundamental limitations in addressing complex SSA tasks. Each mainstream approach offers distinct advantages while facing severe challenges.

Model-based inversion advantages—physical interpretability and result transparency—simultaneously constitute their limitations through excessive physical model dependency:

Extreme Prior Knowledge Dependency: These methods fundamentally presuppose access to precise target three-dimensional geometric models and surface material parameters. For the vast majority of non-cooperative targets, such precise prior information remains strictly classified or fundamentally unobtainable, rendering these methods operationally impractical.

Computational Complexity-Real-time Conflict: Ensuring inversion accuracy requires high-fidelity physical models whose forward simulation demands enormous computation; simultaneously, inversion requires thousands of iterative optimizations, incurring prohibitive costs for real-time or near-real-time SSA requirements. Despite graphics acceleration attempts, the accuracy-speed balance remains elusive for complex attitude motions and fine materials.

Solution Uniqueness and Ill-posed Inversion: Current inversion research attempts to recover multiple parameters (attitude, angular velocity, shape) from limited photometric observations—a fundamentally ill-posed problem where different parameter combinations can produce nearly identical light curves. Optimization algorithms readily fall into local optima, yielding non-unique or completely erroneous inversion results, yet existing research lacks robustness in addressing this challenge.

Feature-based inversion approaches, while circumventing precise physical model dependency, introduce new challenges through “black-box” characteristics and data dependencies:

Feature Representation Limitations: Whether traditional spectral analysis or machine learning-extracted features, most approaches focus on macroscopic parameters like rotation periods and stable states. Current feature extraction methods lack capability for fine shape details or complex non-periodic attitude maneuvers, precluding high-precision shape reconstruction or continuous attitude tracking.

Generalization Vulnerability: A fatal weakness—generalization capability—plagues these approaches. Performance critically depends on training data coverage; encountering novel target types, special materials, or extreme motion patterns absent from training sets triggers catastrophic performance degradation. “Black-box” opacity compounds this vulnerability by rendering performance deterioration unpredictable and inexplicable, posing unacceptable risks in aerospace applications demanding exceptional reliability.

Physical Interpretability Deficit: Purely data-driven models may learn spurious correlations and produce predictions violating fundamental conservation laws. Embedding basic physical constraints within data-driven frameworks to guide learning toward physically plausible results constitutes a critical challenge for next-generation intelligent analysis.

Future research should pursue deep integration rather than parallel coexistence of these trajectories. Physics-guided machine learning, embedding physical models as constraints or priors within neural networks, preserves powerful function approximation capabilities while ensuring physical consistency. Multi-source information fusion combining photometric data with radar, infrared, and other modalities offers pathways to overcome single-source limitations.

4. Intelligent Analysis of Space Object Optical Characteristics

4.1. Traditional Machine Learning Approaches

Machine learning-based inversion represents an emerging paradigm addressing the limitations of traditional methods requiring prior target reflection characteristics and three-dimensional modeling for scattering computation [54]. For unknown non-cooperative targets where traditional approaches face fundamental constraints, machine learning offers pathways to learn target characteristics from historical photometric data and describe them within multi-dimensional feature spaces, enabling real-time anomaly detection when current observations become available. Operational systems based on these principles have been developed and deployed, demonstrating capabilities in space object classification, anomaly detection, feature inversion, and collision warning.

Gong et al. [55] provided a comprehensive review of intelligent space object recognition, positioning, and orbit determination based on photoelectric detection data. Their survey highlights the growing research momentum in this field and underscores the need for automated analysis capabilities as observational capabilities continue to expand.

The following subsections systematically review major technical paradigms in this rapidly evolving field, spanning traditional feature engineering approaches, deep learning architectures, and emerging hybrid methodologies.

4.1.1. Feature Engineering with Classical Classifiers

This most classic and widely applied paradigm extracts discriminative feature vectors from raw light curves through signal processing, statistical analysis, or compressed sensing, and then trains classical machine learning classifiers for recognition. Performance critically depends on the quality of feature engineering, offering advantages of technical maturity, computational efficiency, and interpretability.

The ExoAnalytic Global Telescope Network (EGTN), acquiring approximately 10,000 observations nightly since 2012 with complete GEO belt coverage, provides massive photometric datasets. Lane et al. [56] leveraged this data for Autonomous Characterization Algorithms for Change Detection and Characterization (ACDC) research, systematically investigating machine learning applications in photometric anomaly detection and target classification. Comparing multiple supervised learning algorithms (support vector machines, neural networks, random forests), random forests demonstrated optimal performance across most tasks, particularly achieving extremely high accuracy for prior-known active satellite identification. The process diagram of machine learning and recognition using ACDC is provided in Figure 5.

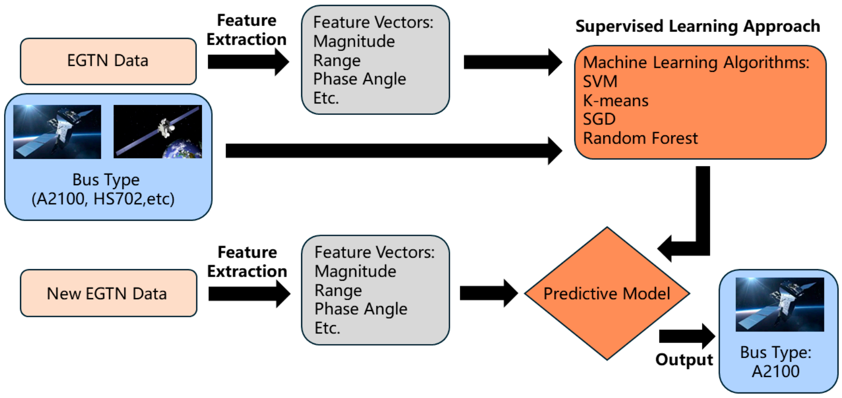

Figure 5. Process diagram of machine learning and recognition using ACDC. Orange: algorithms and model; Gray: feature vectors; Flesh: data; Blue: bus type.

Introducing satellite 'fingerprint' concepts based on annual photometric data [42] provides novel technical pathways for unique satellite identification and status change detection. However, significantly lower classification accuracy for non-active targets reveals limitations when handling chaotic behavioral patterns, suggesting future needs for physics-driven advanced feature extraction to enhance generalization.

Singh [57] developed Athena, a data-driven system using compressed sensing-based sparse feature extraction for robust key feature identification from noisy data. Novel point-wise error-constrained robust principal component analysis enables unsupervised anomaly detection without prior anomaly templates, while supervised learning supports object classification [57]. Athena intelligently collects historical photometric data, learns diverse space object characteristics, and automatically detects anomalous behaviors from massive simulated and measured datasets, forming end-to-end analytical frameworks.

Beyond measured data-driven approaches, simulation-augmented training strategies offer another promising pathway to address the challenge of limited real-world observations. Gartley et al. [58] leveraged the digital imaging and remote sensing image generation (DIRSIG) scene simulation environment to generate high-fidelity hyperspectral signatures for nine classes of unresolved resident space objects. To further enhance classification robustness across diverse operational conditions, they developed a deep learning framework combining multilayer perceptrons with synthetic minority over-sampling technique (SMOTE) and conditional variational autoencoders (CVAE). This work demonstrates the effectiveness of integrating physics-based simulation with data augmentation techniques for overcoming data scarcity in hyperspectral RSO characterization.

Extending the feature engineering paradigm beyond single-band analysis, Macmanus et al. [59] developed a machine learning framework for dual-band characterization of low-Earth orbit satellites using photometric observations. By extracting features from simultaneous visible and near-infrared light curves, their approach enables discrimination of satellite bus types and assessment of material composition. The dual-band methodology provides additional discriminative power compared to single-band analysis, demonstrating that multi-wavelength observations can significantly enhance the information extracted from unresolved photometric measurements for RSO classification.

Beyond methodological developments in feature extraction and data modalities, the operational viability of fingerprint-based characterization depends critically on large-scale systematic validation. Using extensive observational data from the advanced Maui optical and space surveillance (AMOS) network, a large-scale evaluation of photometric fingerprinting and change detection techniques was conducted [60]. This work systematically assessed the stability and discriminative power of satellite light curve fingerprints over time, quantifying how factors such as observational geometry and temporal baseline affect fingerprint reliability. Importantly, automated change detection algorithms were shown to reliably identify configuration changes and anomalies when applied at scale, providing critical validation for transitioning fingerprint-based characterization methods from concept to operational deployment in real-world SSA systems.

4.1.2. Probabilistic Graphical Models with Physical Rules

These approaches abstract photometric variations and underlying physical laws (orbital dynamics, attitude kinematics, optics) into probabilistic models, performing uncertainty-aware inference through graphical models. Core innovations embed domain knowledge as prior probabilities or state transition relationships, achieving interpretable fusion of data-driven and physical mechanism approaches.

GEO satellites frequently suffer misidentification due to co-location and orbital uncertainty. Traditional orbit-based association methods yield high mislabeling rates, while photometric measurements providing physical feature information offer discrimination potential. However, existing methods operating in batch mode introduce high latency and lack real-time capability. Dao [61] addressed this through hidden Markov model-based automated algorithms detecting configuration changes and misidentification in three-axis stabilized GEO satellites. Using differential magnitudes and color indices within Bayesian network (BN) frameworks enabled real-time inference of correct or mislabeled states, with expectation-maximization [62] supporting online learning and parameter updating. This resolved misidentification challenges arising from orbit co-location and uncertainty.

Extending this work, Dao [63] developed COSTB (Concurrent Spatio-Temporal and Brightness Technique), a statistical framework that processes spatiotemporal and photometric information concurrently. COSTB's online operation capability effectively detects intruding non-resident objects by breaking through the traditional separation of position association and photometric analysis. Computing position residual p-values quantifies the likelihood of belonging, enabling a comprehensive determination through likelihood ratio principles for enhanced recognition reliability in complex scenarios such as satellite clusters. However, the establishment of the photometric reference model depends on the quality of historical data, and automated mechanisms for updating seasonal variations remain imperfect, warranting further validation on diverse datasets. A flowchart of COSTB processing is presented in Figure 6.

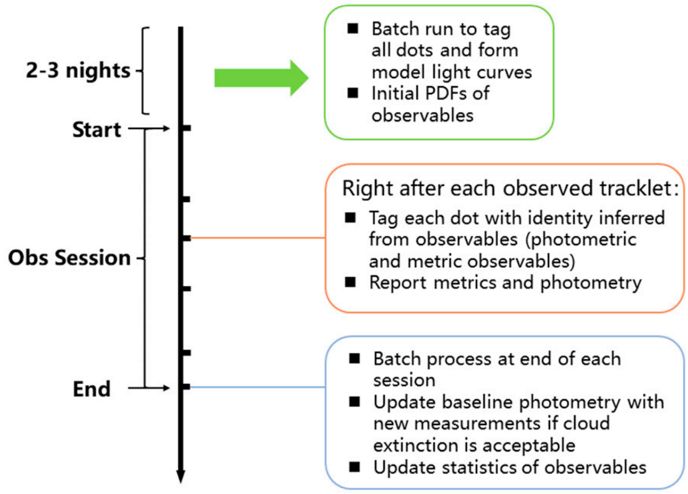

Figure 6. Flowchart of COSTB processing. Colors differentiate the three processing phases (green: initialization; orange: online per-tracklet; blue: batch update).

4.1.3. Optimization for Real-World Data Challenges

These approaches address severe challenges inherent in real-world data: extreme class imbalance, high-dimensional feature redundancy, noise, and missing data. Rather than proposing novel learning algorithms, they introduce cost-sensitive learning, data augmentation, feature selection, and dimensionality reduction to enhance robustness in complex operational scenarios.

The first systematic application of machine learning classification to real non-resolved light curve data for GEO objects addressed challenges including noise, irregular observation intervals, and severe class imbalance [64]. Two strategies were employed: feature extraction using third-order polynomial fitting with coefficients and goodness-of-fit as classification features; and cost-sensitive random forests that adjusted misclassification costs to handle imbalance, significantly enhancing recognition of minority classes (rocket bodies, debris) [64]. Experimental results demonstrated superior performance compared to traditional dynamic time warping-based k-nearest neighbors, establishing the feasibility of machine learning for real SSA tasks and providing practical foundations for automated space object classification [64].

Similar investigations have been conducted using real light curves across a large number of space objects [65]. From tens of thousands of light curves covering 747 objects, 53-dimensional features were extracted using the feats (feature extractor for time series) tool [66]. To mitigate class imbalance, the SMOTE was applied [67]. Principal component analysis (PCA) dimensionality reduction was also evaluated with multiple classical classifiers. While SMOTE significantly enhanced minority class recognition, PCA retaining 99.99% of variance unexpectedly degraded performance, revealing the limitations of linear dimensionality reduction in this feature space [65]. This work provides empirical foundations and methodological guidance for advancing SSA through real photoelectric data.

A uniquely hybrid approach combining knowledge bases with logical reasoning diverges from statistical learning trajectories. Liu et al. [68] proposed a novel integration of ontology with traditional machine learning to address challenges of incomplete observational data. Their OntoStar hybrid knowledge ontology integrated conceptual hierarchies, internal attributes, contextual information, feature derivation rules, and classification rules extracted by the C4.5 decision tree, enabling classification based on logical reasoning. However, ontology-based reasoning efficiency proved significantly slower than traditional methods, limiting its potential for real-time applications.

4.2. Deep Learning Approaches

While traditional machine learning achieved notable success, performance fundamentally depends on the quality of manual feature engineering—researchers manually constructing discriminative features using prior knowledge. However, space object light curves encode complex nonlinear dynamic patterns determined by geometry, materials, and attitude interactions, whose deep feature extraction challenges manual design. As observational data proliferates, traditional representation capabilities approach inherent limits.

Al-Hourani and Manzoor [69] investigated the application of wide field-of-view sensors for optical space situational awareness, addressing the challenge of monitoring an increasingly congested orbital environment. Wide field-of-view systems generate substantial volumes of photometric data, creating both opportunities and demands for automated analysis techniques capable of extracting meaningful information from large-scale observations. This work underscores the growing need for data-driven approaches—such as the deep learning methods reviewed in this section—to process and interpret the rich but complex data streams produced by next-generation optical sensors.

In response to these demands, deep learning technologies enabling automatic hierarchical feature learning from raw or lightly preprocessed data naturally emerge as the next paradigm driving field advancement.

4.2.1. Convolutional Neural Networks

Convolutional neural networks (CNNs), architectures specialized for grid-structured data (images, time series), extract local features through convolutional kernel sliding windows, progressively combining local patterns into complex global representations through multi-layer stacking.

For light curve analysis, treating photometric sequences as one-dimensional time series enables CNNs to effectively capture local fluctuations, specific flare patterns, and brightness trends—local features often associated with specific target structures, postures or motions. CNN-based methods thus excel at learning discriminative features directly related to target identity and status without manual feature design.

Early applications of CNNs to space object classification were enabled by combining the Ashikhmin–Shirley BRDF [70] with rotational dynamics models. Through physical simulation, this approach generated a large number of labeled simulated light curves, which were then used to train CNNs for automatic feature extraction and object classification. While feasibility was demonstrated, model interpretability remained limited; the learned features resisted intuitive understanding, particularly the meanings of high-level convolution kernels. Moreover, training computational costs proved substantial, requiring extensive simulated data generation and computing resources [71].

Extending this work, Linares et al. [72] developed one-dimensional CNN approaches. Large-scale simulated light curve generation spanning diverse shapes, materials, and pose control states addressed the scarcity of real labeled data. Specialized one-dimensional convolutional neural network (1D-CNN) architectures automatically learned discriminative hierarchical features from time series. Systematic comparison with bagged trees [73,74], random forests [75], and support vector machines [76] demonstrated CNN superiority, particularly on real observations, leveraging powerful nonlinear feature extraction for significantly enhanced classification performance. Triplet-distributed stochastic neighbor embedding visualization provided intuitive interpretability for high-dimensional light curve data, revealing class separability characteristics. This work establishes promising deep learning pathways for SSA object recognition.

Deep learning classification performance depends not only on architecture selection but fundamentally on training data scale and labeling accuracy. Allworth et al. [77] addressed real light curve scarcity and labeling challenges through transfer learning-based classification. Blender 3D modeling with physical ray-tracing generated realistic training data, first systematically applying "simulation-to-real" transfer learning frameworks to space object shape classification. Pre-training 1D-CNN on large-scale simulated data followed by fine-tuning on limited real observations demonstrated significant superiority over traditional machine learning performance on both simulated and real datasets, with transfer learning enhancing small-sample classification accuracy. This work provides viable paradigms for deep learning application in data-scarce space object recognition, demonstrating simulated data prior knowledge can significantly reduce real dataset requirements.

The effectiveness of deep neural networks for classifying space objects using light curve time series has also been systematically investigated. By comparing traditional dynamic time warping (DTW) k-nearest neighbors (k-NN) with CNN and recurrent neural network architectures, an innovative conversion of one-dimensional light curves into two-dimensional recurrence plots—used as CNN inputs—expanded the representation of time series features, offering new perspectives for classification [78]. This work establishes a solid technical foundation and feasible research directions for applying deep learning in space surveillance.

Beyond innovations in input representation, the robustness of deep learning approaches under challenging observational conditions warrants systematic investigation. A systematic comparison of multiple machine learning algorithms—including k-NN, random forest, XGBoost, and one-dimensional CNN—has been conducted for classifying single-photon space debris light curves [79]. Operating under the extremely low signal-to-noise conditions characteristic of photon-counting detectors, CNN architectures with automated feature extraction pipelines achieved classification accuracies up to 90.7%, significantly outperforming traditional approaches. This validates the effectiveness of deep learning for extracting discriminative features from noise-dominated photometric measurements.

Building upon the progressive deepening of understanding of network architectures, the potential for synergy among different neural network families has been further explored. A systematic comparison of hybrid recurrent-convolutional networks against pure convolutional networks for multi-channel light curve classification tasks reveals that hybrid architectures—which combine temporal dependency capture with local feature extraction—achieve superior performance in discriminating between active and inactive spacecraft [80]. In addition, the practical implications of class imbalance and normalization strategies have been examined, offering methodological guidance for handling non-uniformly distributed observational data in real-world applications. This research fully demonstrates the potential of ensemble approaches that integrate the strengths of different neural network families for enhanced RSO characterization [80].

Beyond temporal and architectural innovations, recent work has extended deep learning-based classification to complementary data modalities. Landon et al. [81] applied convolutional neural networks to unresolved spectral data from the Falcon Telescope Network for discriminating geosynchronous satellites. Their work demonstrates that spectral signatures—capturing material-dependent reflectance variations across wavelengths—provide complementary discriminative information to traditional photometric light curves. By achieving robust classification performance on spectral measurements alone, this study highlights the potential of multi-modal deep learning approaches that could eventually fuse photometric and spectral data for enhanced space object characterization.

Extending this line of inquiry from multispectral to hyperspectral domain, Kirkendall et al. [82] developed a robust deep learning framework for classifying geosynchronous satellites using unresolved hyperspectral signatures. By leveraging the rich spectral information across contiguous wavelength bands, their convolutional neural network architecture captures material-specific reflectance features that are largely invisible to broadband photometric observations. The approach demonstrates enhanced classification robustness under varying illumination conditions and observational geometries, addressing the generalization challenges that plague purely data-driven methods. This work highlights the potential of hyperspectral data as a complementary modality to traditional light curves for reliable space object characterization.

4.2.2. Recurrent Neural Networks

Recurrent neural networks (RNNs), architectures specialized for sequential data, feature inherent "memory" through recurrent connections enabling persistent information flow, combining current inputs with historical context. Long short-term memory (LSTM) [83] and gated recurrent units represent critical variants addressing long-term dependency challenges through sophisticated gating mechanisms.

Light curves fundamentally represent time series where current brightness depends not only on instantaneous observational geometry but also on past motion states. RNNs, particularly LSTM or gated recurrent unit (GRU), excel at understanding dynamic evolution processes—identifying periodic patterns and non-stationary characteristics arising from target spin, precession, or attitude maneuvers—naturally suited for rotation period inference and attitude anomaly recognition.

Jia et al. [78] demonstrated this through a systematic comparison of DTW k-NN with CNN and RNN architectures, establishing the superiority of deep neural networks (DNNs) in space object classification, with both CNN and RNN significantly outperforming traditional methods.

Addressing the sensitivity of traditional LSTM anomaly detection to noise and the lack of key time point attention mechanisms, Li et al. [84] innovatively proposed a temporal-attention LSTM for unsupervised anomaly detection in space object light curves. Temporal-attention long short-term memory (TA-LSTM) uses LSTM to capture temporal dependencies while integrating temporal-attention mechanisms that assign adaptive importance weights to different time steps.

TA-LSTM comprises three core modules: an input module using sliding windows for sequence segmentation; a prediction module combining LSTM with temporal attention for dynamic key time point focus and next-step magnitude prediction; and an anomaly detection module computing prediction errors with adaptive thresholds derived from normal data statistical distributions for anomaly point identification [84]. Figure 7 provides a flowchart of space object light curve anomaly detection using the TA-LSTM method.

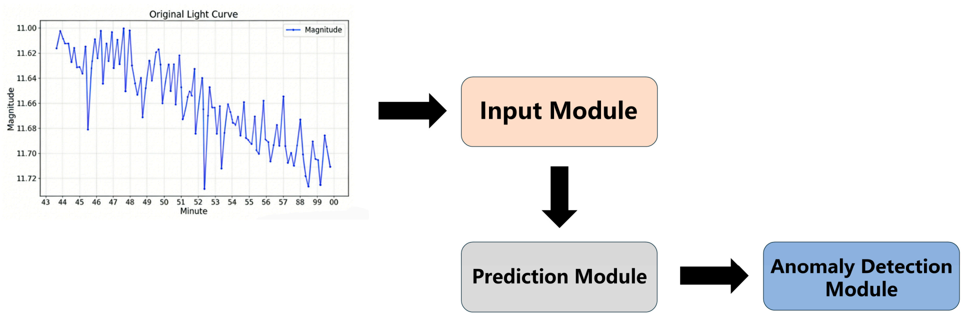

Figure 7. Flowchart of space object light curve anomaly detection using the TA-LSTM method.

Addressing real anomaly labeled data scarcity, Li constructed photometric simulation frameworks based on space object geometry, attitude dynamics, and material reflectance, accurately capturing object–Sun–observer interactions for high-fidelity representation. Experimental results demonstrated TA-LSTM's superiority over mainstream temporal anomaly detection methods, particularly for noise handling and local anomaly detection, providing effective technical pathways for enhanced SSA. However, anomalies with magnitude variations below 0.1 Mv risk being masked by observational noise, potentially causing missed detection or false alarms in real systems.

Beyond direct applications to space objects, methodological insights from adjacent astronomical domains offer valuable transferable knowledge. A hierarchical deep-learning classifier for astronomical transient and variable phenomena, named ORACLE, employs GRUs to achieve real-time classification with as little as a single photometric observation [85]. Its architecture, which concatenates contextual information with light curve embeddings, demonstrates how recurrent networks can extract meaningful features from sparse temporal data—methodological insights directly transferable to RSO characterization under limited observational sequences. This work underscores the broader potential of recurrent architectures for space object analysis, particularly in scenarios where observational opportunities are constrained by sensor availability or target visibility.

4.2.3. Alternative Neural Network Approaches

Beyond CNN and RNN mainstream architectures, alternative neural network models offer unique technical trajectories. Extreme learning machines (ELMs) [86] exemplify this category, representing highly efficient learning algorithms for single-hidden-layer feedforward networks. Core characteristics: random hidden node weights and biases remain fixed without iterative adjustment; output weights determined through one-time linear system solving.

ELM's extreme training speed advantages suit rapid-response scenarios or resource-constrained platforms, providing lightweight efficient nonlinear mapping tools for fast light curve classification or regression. However, representation capacity and generalization typically lag behind deeply optimized CNN or RNN.

Linares and Furfaro [87] developed end-to-end frameworks combining ELM with ontology-based Bayesian networks before their CNN work [70,72]. Three components included: ELM learning energy and momentum parameters from simulated data, enabling rapid measurement-to-physical-parameter mapping for near-real-time inference; clustering analysis in extracted feature spaces identifying behavioral patterns for CNN-based classification; and Space Object Behavior Ontology capturing domain knowledge with semi-automatic Bayesian network transformation for probabilistic inference under uncertainty. This system integrates orbital parameters and operator reports, avoiding purely data-driven or exclusively expert-dependent limitations, supporting high-level collision threat assessment and compliance judgments. However, orbital dynamics employed simplified Newtonian gravity, and Bayesian network conditional probability tables relied on expert-assigned values retaining subjectivity.

4.3. Conclusions and Future Directions

This survey reveals field evolution from simple algorithm application toward interdisciplinary synthesis integrating data-driven and physical-mechanism thinking, offering numerous insights for subsequent research. However, pathways to engineering practicalization confront persistent challenges.

Traditional machine learning offers technical maturity with relatively simple models, computational efficiency, and certain interpretability. Approaches achieving high performance while considering deployment costs and analyst trust embody engineering principles of "practicality over complexity." Yet inherent limitations increasingly constrain further advancement:

Feature Quality Dependency: Performance critically depends on the quality of manually crafted features—phase angle-magnitude curves, statistical features, etc. These features prove inadequate for capturing unknown targets, discriminating complex non-periodic motions, or detecting subtle anomalies; encountering target behaviors beyond prior knowledge triggers rapid performance degradation.

Shallow Learning Capacity Limitations: Traditional models fundamentally represent shallow learning, with insufficient capacity for highly nonlinear relationships arising from high-dynamic postures, complex materials, and observational geometry interactions, limiting achievable recognition accuracy.

Deep learning methods demonstrate end-to-end feature learning potential. CNN and RNN models exhibit powerful automatic feature extraction from raw photometric data, avoiding tedious manual feature engineering and revealing future intelligent analysis directions. However, deep networks carry inherent limitations.

Labeled Data Scarcity Compromising Generalization: Deep learning power demands massive data, critically depending on massive high-quality labeled data—revealing the current bottleneck of "data scarcity." Researchers commonly employ high-fidelity simulated data for training, yet simulation simplifications or errors systematically bias trained models, yielding poor real data performance and fragile generalization.

"Black-Box" Decision Making Creating Trust Crisis: Deep learning models operate as black boxes lacking transparency and interpretability. In high-risk SSA decision support contexts, the inability to explain 'why the target was identified as A rather than B' represents a fatal weakness, severely impeding operational acceptance. Reliability must presuppose interpretability.

Physical Guidance Deficiency: Purely data-driven models may learn spurious correlations, producing predictions that violate fundamental physical laws. Embedding physical knowledge as constraints within neural networks—constructing physics-guided machine learning—constitutes a critical next-generation reliable intelligent analysis.

Additionally, the field lacks public standard benchmark datasets and unified evaluation metrics, hindering fair comparison of different research results. Simultaneously, ground truth parameter unavailability for on-orbit targets complicates final algorithm performance and engineering practicality validation.

Future directions should pursue the following: integration of physical models with data-driven approaches, leveraging physical laws for enhanced generalization and interpretability; explainable AI achieving deep learning performance with interpretability, addressing trust crises; and lightweight model design satisfying real-time processing requirements for operational deployment.

5. Discussion and Future Directions

5.1. Synthesis of Current State

This comprehensive review reveals that space object photometric prediction has evolved through distinct paradigmatic stages. The foundational era established physical understanding through BRDF modeling and OCS computation, providing rigorous frameworks linking material properties to observable signatures. The inversion era developed methodologies for extracting target characteristics from photometric measurements, encompassing both model-based optimization and feature-based pattern recognition. The current intelligent analysis era leverages machine learning and deep learning to automatically discover complex mappings from data, offering unprecedented predictive capabilities.

Each paradigm contributes essential insights while exhibiting characteristic limitations. Physical models provide interpretability and generalizability grounded in conservation laws but require extensive prior knowledge and incur computational costs. Feature-based inversion offers prior-free operation but captures only macroscopic characteristics. Deep learning shows superior predictive accuracy in existing comparative studies on their respective datasets, but the lack of common benchmarks makes it difficult to assert universal superiority; moreover, it struggles with generalization and interpretability. The field's maturity now demands synthesis rather than continued parallel evolution.

5.2. Critical Research Gaps

Simulation-to-Reality Gap: The scarcity of real labeled photometric data forces reliance on simulated training datasets. However, inevitable simplifications in BRDF models, geometric representations, and environmental factors introduce systematic discrepancies between simulated and real measurements. Models trained exclusively on simulation exhibit degraded performance when deployed operationally. Bridging this gap requires either higher fidelity simulation incorporating space environmental effects, or domain adaptation techniques aligning simulated and real feature distributions.

Interpretability–Accuracy Trade-off: Deep learning achieves state-of-the-art predictive accuracy but operates opaquely. For high-consequence SSA decisions, unexplained predictions remain unacceptable. Conversely, interpretable physical models sacrifice accuracy for transparency. Developing architectures that maintain deep learning's representational power while providing meaningful explanations constitutes a fundamental challenge. Promising templates for addressing this trade-off emerge from adjacent fields. In the context of multi-angle remote sensing, Liu et al. [88] proposed a BRDF-based spectral curve inversion model that fits reflectance variation across detection angles using a kernel-driven BRDF formulation. By decomposing spectral curves into individual wavebands and modeling their angular dependence, their approach successfully reconstructs angle-consistent spectra with coefficients of determination exceeding 0.83. While developed for terrestrial target detection, this methodology offers a template for integrating BRDF physical constraints into data-driven inversion frameworks for space object characterization—a hybrid paradigm that preserves physical interpretability while leveraging the flexibility of data-driven optimization.

Temporal Multi-Scale Modeling: Photometric signatures encode information across multiple temporal scales: sub-second flashes from specular facets, seconds-to-minutes variations from rotation, hours-to-days evolution from attitude precession, and months-to-years changes from orbital motion and environmental degradation. Current models typically address single scales, lacking unified frameworks capturing cross-scale interactions.

Benchmarking Infrastructure Deficit: Unlike established machine learning domains with standardized datasets and evaluation protocols, space object photometric analysis lacks public benchmarks. This impedes progress by preventing fair comparison, obscuring relative advantages of different approaches, and discouraging entry by researchers outside the immediate field.

Cross-Domain Methodological Transfer: Beyond the immediate field of space object characterization, valuable methodological insights can be drawn from adjacent domains facing similar data challenges. El-Borady [89] provided a comprehensive review of machine learning applications in stellar variability research, covering the full pipeline from data cleaning to classification and parameter inference using large-scale photometric surveys. The systematic treatment of representation learning, multimodal datasets, and foundation models for time-domain astronomy offers transferable methodologies for RSO light curve analysis, particularly given the shared challenges of irregular sampling, systematic noise, and class imbalance. Such cross-disciplinary fertilization represents an underexplored opportunity for advancing space object photometric prediction capabilities—one that could accelerate progress by leveraging decades of methodological development in astrophysics.

A particularly transferable technique comes from Fiscale et al. [90], who developed DART-Vetter, a compact convolutional neural network that processes period-folded light curves to distinguish planetary candidates from false positives in transit survey data. Despite its simple architecture, the model achieves highly competitive triaging performance with 91% recall on combined TESS and Kepler datasets, demonstrating that well-designed CNNs can extract discriminative features from phase-folded photometric sequences. This approach offers direct methodological parallels for RSO classification tasks.

Polarimetric Measurements as a Complementary Modality: Beyond the methodological gaps discussed above, polarimetric light curves represent an emerging observational dimension that offers additional insights into surface material properties and scattering geometry. Unlike total-intensity photometry, polarization encodes information about the surface roughness, composition, and structure of the reflecting facets. For geosynchronous satellites, polarimetric signatures—specifically the degree of linear polarization (DOLP) and the angle of linear polarization (AOLP)—have been collected using a linear polarimeter mounted to a 16-inch telescope, demonstrating utility for space object identification and characterization [91]. Such measurements can help discriminate between different satellite surface materials and break degeneracies that arise from intensity-only observations. While not covered in detail here due to scope and space limitations, polarimetric analysis is worthy of further investigation in future work as a complement to conventional photometric methods.

6. Conclusions

This review has comprehensively examined data-driven approaches for space object photometric prediction, synthesizing research across optical scattering characterization, shape and attitude inversion, and intelligent analysis methodologies. The field has progressed from foundational physical models through inversion techniques to contemporary machine learning applications, each contributing essential capabilities while revealing characteristic limitations.

Optical scattering research established BRDF frameworks linking material properties to observable signatures, with OCS computation enabling integrated target characterization. Shape and attitude inversion developed both model-based optimization and feature-based pattern recognition approaches for extracting target information from light curves. Intelligent analysis introduced machine learning's automatic feature discovery, achieving unprecedented predictive accuracy while confronting interpretability and generalization challenges.

Critical analysis identifies persistent gaps: the simulation-reality divide limiting operational deployment, the interpretability–accuracy trade-off impeding trust, temporal multiscale modeling deficiencies, and benchmarking infrastructure absence. Future progress requires principled synthesis through physics-guided machine learning, explainable AI tailored for photometric analysis, multi-modal information fusion, generative data augmentation, and lifelong learning frameworks.

As space objects proliferate and the orbital environment grows increasingly congested, robust photometric prediction capabilities become ever more essential for collision avoidance, anomaly detection, and space traffic management. The research directions charted herein provide pathways toward next-generation space situational awareness systems combining physical rigor with data-driven adaptability, interpretability with predictive accuracy, and laboratory validation with operational reliability.

Author Contributions: Conceptualization, L.Y., Y.Z. and C.X.; resources, L.Y.; writing—original draft preparation, L.Y.; writing—review and editing, L.Y., Y.Z. and C.X.; supervision, Y.Z. and C.X.; funding acquisition, Y.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Acknowledgments: During the preparation of this manuscript, the authors used a large language model (DeepSeek, version 2026-03) for the purposes of English language editing and manuscript formatting assistance. The authors have reviewed and edited all AI-generated content and take full responsibility for the final published version.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

1D-CNN	One-Dimensional Convolutional Neural Network
ACDC	Autonomous Characterization Algorithms for Change Detection and Characterization
AMOS	the Advanced Maui Optical and Space Surveillance
AOLP	the Angle of Linear Polarization
BN	Bayesian Network
BRDF	Bidirectional Reflectance Distribution Function
CCD	Charge-Coupled Device
CHEOPS	CHaracterising ExOPlanets Satellite
CMOS	Complementary Metal-Oxide-Semiconductor
CNN	Convolutional Neural Network
COSTB	Concurrent Spatio-Temporal and Brightness Technique
CVAE	Conditional Variational Autoencoder
DAMIT	Database of Asteroid Models from Inversion Techniques
DIRSIG	Digital Imaging and Remote Sensing Image Generation
DOLP	the Degree of Linear Polarization
DR3	Data Release 3
DNN	Deep Neural Network
DTW	Dynamic Time Warping
EGTN	the ExoAnalytic Global Telescope Network
ELM	Extreme Learning Machine
GEO	Geostationary Earth Orbit
GRU	Gated Recurrent Unit
IoU	Intersection over Union
k-NN	k-Nearest Neighbors
LEO	Low Earth Orbit
LSTM	Long Short-Term Memory
MLI	Multi-Layer Insulation
MLP	Multilayer Perceptron
NeRF	Neural Radiance Field
NeuMIP	Neural Multi-resolution Material Representation
nsPSO	Non-social Particle Swarm Optimization
OCS	the Optical Cross Section
OpenGL	Open Graphics Library
PCA	Principal Component Analysis
PSO	Particle Swarm Optimization
RGB	Red Green Blue
RNN	Recurrent Neural Network
RPV	Rahman–Pinty–Verstraete
RSO	Resident Space Object

RSONAR	Resident Space Object Network for Assessment and Recognition
SMOTE	Synthetic Minority Over-sampling Technique
SSA	Space Situational Awareness
STK	Satellite Tool Kit
TA-LSTM	Temporal-Attention Long Short-Term Memory
TESS	Transiting Exoplanet Survey Satellite
VESPA	Vega Secondary Payload Adapter
XGBoost	eXtreme Gradient Boosting

References

1. Frueh, C.; Howell, K.; DeMars, K.J.; Bhadauria, S. Cislunar Space Situational Awareness. In Proceedings of the 31st AIAA/AAS Space Flight Mechanics Meeting, Virtual, 1–3 February 2021.
2. Fiedler, H.; Bergmann, C.; Griese, F.; Herzog, J.; Hofmann, B.; Kleint, L.; Zollo, A. Five Years of SMARTnet: Data, Processing, and Improvements. In Proceedings of the 2nd ESA NEO and Debris Detection Conference, Darmstadt, Germany, 24–26 January 2023.
3. Goumiri, I.; Muyskens, A.; Priest, B.; Armstrong, R. Light Curve Forecasting and Anomaly Detection Using Scalable, Anisotropic, and Heteroscedastic Gaussian Process Models. In Proceedings of the Advanced Maui Optical and Space Surveillance (AMOS) Technologies Conference, Wailea, HI, USA, 19–22 September 2023.
4. Vananti, A.; Lu, Y.; Schildknecht, T. Attitude Estimation of H2A Rocket Body from Light Curve Measurements. *Int. J. Astrophys. Space Sci.* **2023**, *11*, 15–22. [CrossRef]
5. Sridharan, R.; Pensa, A.F. U.S. Space Surveillance Network Capabilities. *Proc. SPIE* **1998**, *3434*, 88–100.
6. Nicodemus, F.E. Directional Reflectance and Emissivity of an Opaque Surface. *Appl. Opt.* **1965**, *4*, 767–775. [CrossRef]
7. Phong, B.T. Illumination for Computer Generated Pictures. *Commun. ACM* **1975**, *18*, 311–317. [CrossRef]
8. Wu, Z.S.; Xie, D.H.; Xie, P.H.; Wei, Q.N. Genetic Algorithm for Statistical Modeling of Rough Surface Laser Scattering. *Acta Opt. Sin.* **2002**, *22*, 897–901.
9. Liu, C.H.; Li, Z.; Xu, C. A Modified Phong Model for Fresnel Reflection Phenomenon of Commonly Used Materials for Space Targets. *Laser Optoelectron. Prog.* **2017**, *54*, 102901. [CrossRef]
10. Torrance, K.E.; Sparrow, E.M. Theory for Off-Specular Reflection from Roughened Surfaces. *J. Opt. Soc. Am.* **1967**, *57*, 1105–1114. [CrossRef]
11. Cook, R.L.; Torrance, K.E. A Reflectance Model for Computer Graphics. *ACM Trans. Graph.* **1982**, *1*, 7–24. [CrossRef]
12. Schlick, C. An Inexpensive BRDF Model for Physically-Based Rendering. *Comput. Graph. Forum.* **1994**, *13*, 233–246. [CrossRef]
13. Lafortune, E.P.F.; Foo, S.C.; Torrance, K.E.; Greenberg, D.P. Non-Linear Approximation of Reflectance Functions. In Proceedings of the 24th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '97), Los Angeles, CA, USA, 3–8 August 1997.
14. Donnelly, K.P. Macroscopic Roughness Modeling of Satellite Multi-Layer Insulation Reflectance. Master's Thesis, Rochester Institute of Technology, Rochester, NY, USA, 2025.
15. Kuznetsov, A.; Mullia, K.; Xu, Z.; Hašan, M.; Ramamoorthi, R. NeuMIP: Multi-Resolution Neural Materials. *ACM Trans. Graph.* **2021**, *40*, 175. [CrossRef]
16. Zhang, L.; Rupnik, E.; Nguyen, T.D.; Jacquemoud, S.; Klinger, Y. BRDF-NeRF: Neural Radiance Fields with Optical Satellite Images and BRDF Modelling. *Int. J. Appl. Earth Obs. Geoinf.* **2025**, *143*, 104747. [CrossRef]
17. Bao, W.Z.; Cong, M.Y.; Zhang, W.; Cheng, J.; Cao, Y.M. Computation Method for Optical Characteristics of Space Objects Based on Facet Discretization. *J. Harbin Inst. Technol.* **2010**, *42*, 710–715.
18. Han, Y.; Sun, H.; Li, Y.; Guo, H. Fast Calculation Method of Complex Space Targets' Optical Cross Section. *Appl. Opt.* **2013**, *52*, 4013. [CrossRef]
19. Xu, C.; Zhang, Y.S.; Li, P.; Li, J.L. Calculation of Optical Cross Section Areas of Spatial Objects Based on OpenGL Picking Technique. *Acta Opt. Sin.* **2017**, *37*, 0720001. [CrossRef]
20. Maxwell, J. Efficient Photometric Modeling of Complex Spacecraft Geometries toward Next-Generation Light Curve Analysis. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
21. Tian, Q.C.; Li, Z.; Xu, C.; Li, P.; Liu, C.H.; Qi, Y. Comparative Validation of Satellite Optical Scattering Characteristics Based on Experimental Measurement and OCS Simulation. *Acta Photonica Sin.* **2018**, *47*, 0129004. [CrossRef]
22. Hadji Hossein, S.; Cimino, L.; Rossetti, M.; Zarcone, G.; Mariani, L.; Curianò, F.; Bucciarelli, M.; Seitzer, P.; Santoni, F.; Di Cecco, A.; et al. Photometric Characterization of Starlink Satellite Tracklets Using RGB Filters. *Adv. Space Res.* **2023**, *72*, 2741–2758. [CrossRef]

23. Billot, N.P.; Correia, A.C.M.; Courcol, B.; Da Silva, J.C.; Deleuil, M.; Ehrenreich, D.; Heng, K.; Benz, W. The CHEOPS View: Serendipitous Observations of Space Objects. *J. Space Saf. Eng.* **2024**, *11*, 456–467.
24. Suthakar, V.; Porto, I.; Myhre, M.; Sanvido, A.A.; Clark, R.; Lee, R.S.K. RSONAR: Data-Driven Evaluation of Dual-Use Star Tracker for Stratospheric Space Situational Awareness (SSA). *Sensors* **2026**, *26*, 179. [CrossRef]
25. Du, Y.Z. Research on Visible Light Scattering Characteristics of On-Orbit Satellites Based on Component-Based Modeling. Master's Thesis, Xidian University, Xi'an, China, 2019.
26. Zhang, Y.S.; Fan, P.S.; Liu, H.Y. *Mastering and Proficiency in Satellite Tool Kit (STK)*; National Defense Industry Press: Beijing, China, 2011.
27. Han, Y.; Chen, M.; Xie, J.F.; Duan, C.L. Simulation and Validation of Target Spacecraft Imaging by Ground-Based Optical Telescopes. *Infrared Laser Eng.* **2019**, *48*, 1214002. [CrossRef]
28. Calatroni, C.; Métris, G.; Courde, C.; Phung, D.H.; Chabé, J.; Aimar, M.; Maurice, N.; Mariey, H. A New Model to Predict Ajisai Satellite Reflected Sunlight Flashes and Application to the Determination of Its Rotation Parameters. *Adv. Space Res.* **2025**, *75*, 7327–7344. [CrossRef]
29. Li, Z.; Wang, Z.; Yang, J.; Hasan, M.; Wang, B. PureSample: Neural Materials Learned by Sampling Microgeome-try. *arXiv* **2025**, arXiv:2508.07240.
30. Wang, Y.; Du, X.; Fan, C. Progress of Light Curve Inversion Technology for Resident Space Object Characteristics. *Chin. Sci. Bull.* **2017**, *62*, 1578–1590. [CrossRef]
31. Wetterer, C.J.; Jah, M. Attitude Determination from Light Curves. *J. Guid. Control Dyn.* **2009**, *32*, 1648–1651. [CrossRef]
32. Clark, R.; Fu, Y.; Dave, S.; Lee, R.S.K. Resident Space Object (RSO) Attitude and Optical Property Estimation from Space-Based Light Curves. *Adv. Space Res.* **2022**, *70*, 3271–3280. [CrossRef]
33. Shan, B.; Liang, Y.Q.; Li, H.N. Attitude and Angular Velocity Estimation of Space Objects Based on Photometric Observations. *Acta Opt. Sin.* **2017**, *37*, 144–155.
34. Renis, F. Leveraging Light-Curve Inversion for Real-Time Kinematic State Estimation of Uncooperative Targets. Master's Thesis, Politecnico di Torino, Turin, Italy, 2025.
35. Tian, Q.C.; Li, Z.; Xu, C.; Li, P.; Liu, C.H.; Qi, Y. Motion Analysis of Precession Space Target Based on Optical Cross Section. *Infrared Laser Eng.* **2017**, *46*, 90–95.
36. Zhang, F.; Li, Y.; Xu, C.; Zhao, S. Multivariate Models of Satellite Optical Scattering Characteristics. *Acta Opt. Sin.* **2020**, *40*, 729001. [CrossRef]
37. Gonçalves, L.F.P.; Simone, N.; de Almeida, A.K.; Barbosa, D.; Jah, M.; Vaillant, T.; Coelho, B.; Correia, A.C.M. Light-Curve Inversion for Anthropogenic Space Objects Using Multi-Colour Constraints. *Acta Astronaut.* **2026**, *241*, 378–408. [CrossRef]
38. Rubio, J.; de Andrés, A.; Paulete, C.; Gallego, Á.; Lozano, J.M.; Escobar, D.; Agueda, A. A High-Performance and Robust Light Curve Inversion Method for Attitude Monitoring of Three-Axis Stabilised Satellites. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
39. Robinson, L.; Frueh, C. A CCD/CMOS Telescope Digital Twin for Space Situational Awareness. *Adv. Space Res.* **2025**, *76*, 3074–3097. [CrossRef]
40. Burton, A.; Robinson, L.; Frueh, C. Light Curve Attitude Estimation Using Particle Swarm Optimizers. *Adv. Space Res.* **2024**, *74*, 5619–5638. [CrossRef]
41. Hall, D.; Calef, B.; Knox, K.; Bolden, M.; Kervin, P. Separating Attitude and Shape Effects for Non-Resolved Objects. In Proceedings of the 2007 AMOS Technical Conference, Wailea, HI, USA, 11–14 September 2007.
42. Tian, Q.C.; Li, Z.; Xu, C.; Li, P. Rotation Analysis of Unstable Space Targets Based on Optical Scattering Characteristics. *Chin. J. Light Scatt.* **2017**, *29*, 266–270.
43. Benson, C.J.; Scheeres, D.J.; Moskovitz, N.A. Spin State Evolution of Asteroid (367943) Duende During Its 2013 Earth Flyby. *Icarus* **2020**, *340*, 113518. [CrossRef]
44. Isoletta, G.; Opromolla, R.; Fasano, G. Attitude Motion Classification of Resident Space Objects Using Light Curve Spectral Analysis. *Adv. Space Res.* **2025**, *75*, 1077–1095. [CrossRef]
45. VanderPlas, J.T. Understanding the Lomb–Scargle Periodogram. *Astrophys. J. Suppl. Ser.* **2018**, *236*, 16. [CrossRef]
46. Stellingwerf, R.F. Period Determination Using Phase Dispersion Minimization. *Astrophys. J.* **1978**, *224*, 953. [CrossRef]
47. Bencivenga, P.; Isoletta, G.; Opromolla, R.; Fasano, G. Attitude Motion Characterization of Resident Space Objects via Fusion of Ground-based and Space-based Light Curves. In Proceedings of the 27th International Conference on Information Fusion (FUSION 2024), Venice, Italy, 8–11 July 2024.
48. Bhowmick, U.; Kumaran, S. Lightcurve Inversion: Analyzing the Extent of Feature Embedding of a Two-Dimensional Projected Shape on Its Transit Lightcurve. In Proceedings of the 43rd Annual Meeting of the Astronomical Society of India, Rourkela, India, 15–19 February 2025.

49. Yoshida, J.; Sano, T.; Taya, K.; Mori, D.; Otagiri, M. Unsupervised Metric Learning for Satellite Attitude Estimation. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
50. Takács, N.; Kiss, C.; Szakáts, R.; Pál, A. Solar System Objects Observed with TESS—Early Data Release 2: I. Spin-Shape Recovery Potential. *Publ. Astron. Soc. Pac.* **2025**, *137*, 044401. [CrossRef]
51. Lim, T.W.; Chun, F.K.; Roth, B.D.; Strong, D.M.; Giblin, T.W. Correlation of Rotational Motion of Rocket Bodies with Falcon Telescope Network Photometric Observations. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
52. Tang, Y.J.; Ying, C.C.; Xia, C.Z.; Zhang, X.M.; Jiang, X.J. Asteroid Shape Inversion with Light Curves Using Deep Learning. *Astron. Astrophys.* **2025**, *696*, A55. [CrossRef]
53. Tsaprailis, K.; Choumos, G.; Kontoes, C.; Lappas, V. Integrating AI in Space Operations: Spin Status Characterization and Initial Orbit Determination from Single Optical Tracks. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
54. Zhang, Y.S.; Xie, X.G.; Su, H.; Zhang, F.Z.; Li, G. Research on Calculation Methods for Optical Scattering Characteristics of Space Objects. *High Power Laser Partical Beams* **2025**, *37*, 099002.
55. Gong, M.Z.; Yang, T.Z.; Chen, H.; Wang, L.; Xia, C.X. A Review of Intelligent Identification, Positioning, and Orbit Determination of Space Targets Based on Photoelectric Detection Data. *Technol. Innov. Appl.* **2025**, *15*, 114–117.
56. Lane, B.; Poole, M. Using Machine Learning for Advanced Anomaly Detection and Classification. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference, Wailea, HI, USA, 20–23 September 2016.
57. Singh, N. Athena: A Data-Driven Anomaly Detection and Space Object Classification Tool for SSA. In Proceedings of the AIAA/AAS Astrodynamics Specialist Conference, Long Beach, CA, USA, 13–16 September 2016.
58. Gartley, M.; Kirkendall, J.; Najera, A.; Velez-Reyes, M. Development of a Robust Deep Learning Model for Classification of Non-Resolved Resident Space Object Hyperspectral Signatures. In Proceedings of the SPIE Defense + Commercial Sensing, Orlando, FL, USA, 13–17 April 2025.
59. Macmanus, C.; Cerff, B.; Lambour, R.; Stutts, L.; Evans, N. Dual-band Characterisation of Low-Earth Orbit Satellites Using Machine Learning Techniques. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
60. Reid, T.; Craychee, T.; Boetcher, A.; Skuljan, J.; Simms, L. Evaluating Photometric Fingerprinting and Change Detection at Scale. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
61. Dao, P. Automated Algorithm to Detect Changes in Geostationary Satellite's Configuration and Cross-Tagging. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference, Wailea, HI, USA, 15–18 September 2015.
62. Dempster, A.P.; Laird, N.M.; Rubin, D.B. Maximum Likelihood from Incomplete Data via the EM Algorithm. *J. R. Stat. Soc. Ser. B Methodol.* **1977**, *39*, 1–22. [CrossRef]
63. Dao, P.; Heinrich-Josties, E.; Boroson, T. Automated Algorithms to Identify Geostationary Satellites and Detect Mistagging Using Concurrent Spatio-Temporal and Brightness Information. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference, Wailea, HI, USA, 20–23 September 2016.
64. Bennette, W.D.; Zeliff, K.; Raquepas, J. Classification of Objects in Geosynchronous Earth Orbit via Light Curve Analysis. In Proceedings of the 2017 IEEE Symposium Series on Computational Intelligence (SSCI), Honolulu, HI, USA, 27 November–1 December 2017.
65. Khalil, M.; Fantino, E.; Liatsis, P. Classification of Space Objects Using Machine Learning Methods. In Proceedings of the 2019 IEEE First International Conference on Cognitive Machine Intelligence (CogMI), Los Angeles, CA, USA, 12–14 December 2019.
66. Cabral, J.B.; Sánchez, B.; Ramos, F.; Gurovich, S.; Granitto, P.M. From FATS to feets: Further improvements to an astronomical feature extraction tool based on machine learning. *Astron. Comput.* **2018**, *25*, 213–220. [CrossRef]
67. Chawla, N.V.; Bowyer, K.W.; Hall, L.O.; Kegelmeyer, W.P. SMOTE: Synthetic Minority Over-Sampling Technique. *J. Artif. Intell. Res.* **2002**, *16*, 321–357. [CrossRef]
68. Liu, B.; Yao, L.; Han, D. Harnessing Ontology and Machine Learning for RSO Classification. *SpringerPlus* **2016**, *5*, 1655. [CrossRef]
69. Al-Hourani, A.; Manzoor, B. Optical-Based Space Situational Awareness Using Wide Field-of-View Sensors. *Acta Astronaut.* **2026**, *238*, 1063–1072. [CrossRef]
70. Ashikhmin, M.; Shirley, P. An Anisotropic Phong BRDF Model. *J. Graph. Tools* **2000**, *5*, 25–32. [CrossRef]
71. Linares, R.; Furfaro, R. Space Object Classification Using Deep Convolutional Neural Networks. In Proceedings of the 19th International Conference on Information Fusion (FUSION), Heidelberg, Germany, 5–8 July 2016.
72. Linares, R.; Furfaro, R.; Reddy, V. Space Objects Classification via Light-Curve Measurements Using Deep Convolutional Neural Networks. *J. Astronaut. Sci.* **2020**, *67*, 1063–1091. [CrossRef]

73. Dietterich, T.G. Ensemble Methods in Machine Learning. In *Multiple Classifier Systems*; Springer: Berlin/Heidelberg, Germany, 2000; pp. 1–15.
74. Quinlan, J.R. Induction of Decision Trees. *Mach. Learn.* **1986**, *1*, 81–106. [CrossRef]
75. Breiman, L. Random Forests. *Mach. Learn.* **2001**, *45*, 5–32. [CrossRef]
76. Hearst, M.A.; Dumais, S.T.; Osuna, E.; Platt, J.; Scholkopf, B. Support Vector Machines. *IEEE Intell. Syst. Their Appl.* **1998**, *13*, 18–28. [CrossRef]
77. Allworth, J.; Windrim, L.; Bennett, J.; Bryant, M. A Transfer Learning Approach to Space Debris Classification Using Observational Light Curve Data. *Acta Astronaut.* **2021**, *181*, 301–315. [CrossRef]
78. Jia, B.; Pham, K.D.; Blasch, E.; Wang, Z.; Shen, D.; Chen, G. Space Object Classification Using Deep Neural Networks. In Proceedings of the 2018 IEEE Aerospace Conference, Big Sky, MT, USA, 3–10 March 2018.
79. Trummer, N.M.; Reza, A.; Steindorfer, M.A.; Helling, C. Machine Learning-Based Classification for Single Photon Space Debris Light Curves. *Acta Astronaut.* **2025**, *226*, 542–554. [CrossRef]
80. Bencivenga, P.; Matacera, M.A.; Isoletta, G.; Opromolla, R.; Fasano, G. Neural Network Analysis of Multi-Channel Light Curves for the Characterization of Resident Space Objects. In Proceedings of the 76th International Astronautical Congress, Sydney, Australia, 29 September–3 October 2025.
81. Landon, G.; Strong, D.; Giblin, T.; Roth, B.; Chun, F. Deep Learning Based Classification of GEOs using Unresolved Spectral Data. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
82. Kirkendall, B.; Krawczyk, B.; Chun, F.; Gartley, M. Leveraging Unresolved Hyperspectral Signatures for Robust Deep Learning Classification of Geosynchronous Satellites. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS) 2025, Maui, HI, USA, 16–19 September 2025.
83. Hochreiter, S.; Schmidhuber, J. Long Short-Term Memory. *Neural Comput.* **1997**, *9*, 1735–1780. [CrossRef]
84. Li, W.; Zhang, Y.; Wang, J.; Liu, X. Detecting Anomalies in Space-Object Light Curves Using Temporal-Attention LSTM. *Space Habitat.* **2025**, *1*, 100007. [CrossRef]
85. Shah, V.G.; Gagliano, A.; Malanchev, K.; Narayan, G.; Malz, A.I.; LSST Dark Energy Science Collaboration. ORACLE: A Real-Time, Hierarchical, Deep-Learning Photometric Classifier for the LSST. *Astrophys. J.* **2025**, *995*, 4. [CrossRef]
86. Huang, G.B.; Zhou, H.; Ding, X.; Zhang, R. Extreme Learning Machine for Regression and Multiclass Classification. *IEEE Trans. Syst. Man Cybern. Part B* **2012**, *42*, 513–529. [CrossRef]
87. Linares, R.; Furfaro, R. Resident Space Object Characterization and Behavior Understanding via Machine Learning and Ontology-Based Bayesian Networks. In Proceedings of the Advanced Maui Optical and Space Surveillance Technologies Conference, Wailea, HI, USA, 20–23 September 2016.
88. Liu, Z.; Sun, X.; Sha, J.; Sun, H.; Wang, D.; Gao, L. A Multi-Angle Spectral Curve Inversion Model Based on BRDF. *Chin. Space Sci. Technol.* **2025**, *45*, 37–47.
89. Audenaert, J. From Stellar Light to Astrophysical Insight: Automating Variable Star Research with Machine Learning. *Astrophys. Space Sci.* **2025**, *370*, 72. [CrossRef]
90. Fiscale, S.; Inno, L.; Rotundi, A.; Ciaramella, A.; Ferone, A.; Magliano, C.; Cacciapuoti, L.; Kostov, V.; Quintana, E.V.; Covone, G.; et al. DART-Vetter: A Deep Learning Tool for Automatic Triage of Exoplanet Candidates. *Astron. J.* **2025**, *170*, 73. [CrossRef]
91. Eastman, B.; Strong, D.; Chun, F.; O'Keefe, D. Linear Polarization Signatures of Geosynchronous Satellites. *J. Astronaut. Sci.* **2024**, *71*, 39. [CrossRef]